SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303
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COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS: **Total Marks: 20**

Question 1: Responsive Navigation Bar Design - Marks(4)
Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.
(a) Identify key components required in a navigation bar for an e-commerce website.
(b) Design a responsive navigation bar using appropriate CSS techniques.
(c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
(d) Evaluate the usability and suggest improvements for better user experience.

Ans:

(a) Key Components

- Website logo
- Navigation links (Home, Products, Categories, Cart, Contact)
- Search bar
- User account/login
- Shopping cart icon

(b) Design

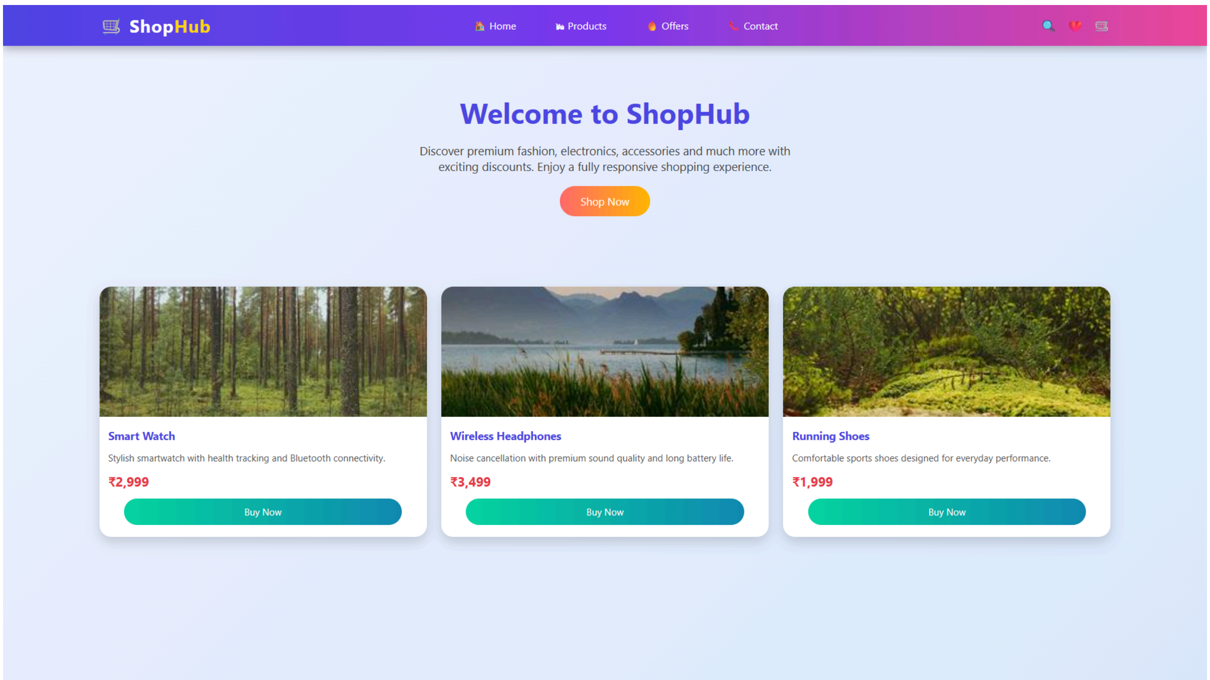
- Use HTML for the navigation structure.
- Use CSS Flexbox for alignment.
- Apply gradient background, hover effects, and spacing.

(c) Responsive Adaptation

- Use media queries for smaller screens.
- Hide navigation links and display a hamburger menu.
- Toggle the menu using JavaScript.

(d) Evaluation

- Responsive on all devices.
- Easy navigation and improved accessibility.
- Smooth animations enhance user experience.



Question 2: Product Hover Effect - Marks(4)
Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.
(a) Identify suitable CSS properties for hover effects.
(b) Suggest appropriate CSS animation or transition for product interaction.
(c) Design a hover effect for a product card.
(d) Evaluate performance and user experience considerations.

(a) CSS Properties

- transform
- transition
- box-shadow
- opacity
- background-color

(b) CSS Animation/Transition

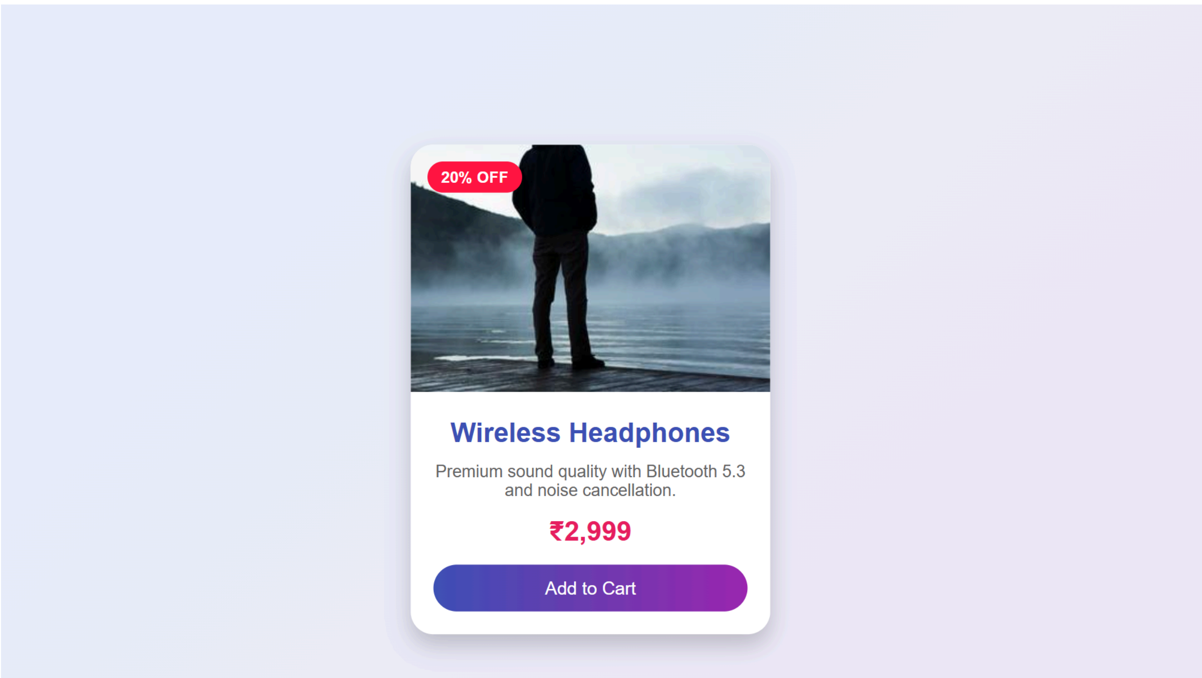
- Use transition: 0.3s ease;
- Apply transform: scale() on hover.
- Add shadow and color transition.

(c) Hover Effect Design

- Zoom the product image.
- Lift the product card.
- Add shadow and highlight the button.

(d) Evaluation

- Makes products more attractive.
- Improves customer interaction.
- Keep animations lightweight for better performance.



Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.
(a) Identify layout requirements of a dashboard (rows, columns, alignment).
(b) Compare Flexbox and Grid based on layout needs.
(c) Choose the most suitable layout technique and justify your choice.
(d) Evaluate the scalability and responsiveness of the chosen approach.

Ans:(a) Layout Requirements

- Header
- Sidebar
- Main content
- Dashboard cards
- Footer

(b) Comparison

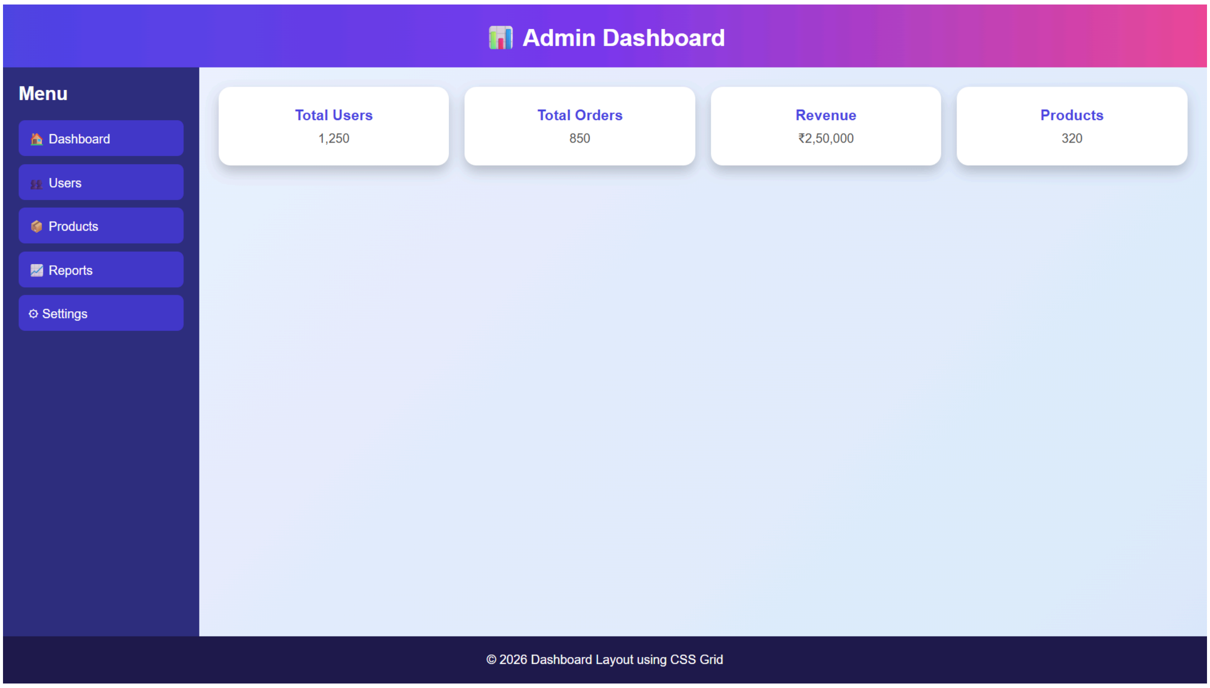
Flexbox	Grid
One-dimensional layout	Two-dimensional layout
Best for rows or columns	Best for complete page layouts
Easy alignment	Better for dashboards

(c) Choice

- CSS Grid is preferred because it efficiently manages rows and columns in dashboard layouts.

(d) Evaluation

- Easy to maintain.
- Highly responsive.
- Suitable for large and complex layouts.



Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.
(a) Identify key sections of a blog layout.
(b) Design a mobile-first layout using CSS.
(c) Demonstrate how the layout scales for larger screens using media queries.
(d) Evaluate accessibility and readability of the design.

Ans:(a) Key Sections

- Header
- Navigation
- Blog posts
- Sidebar
- Footer

(b) Design

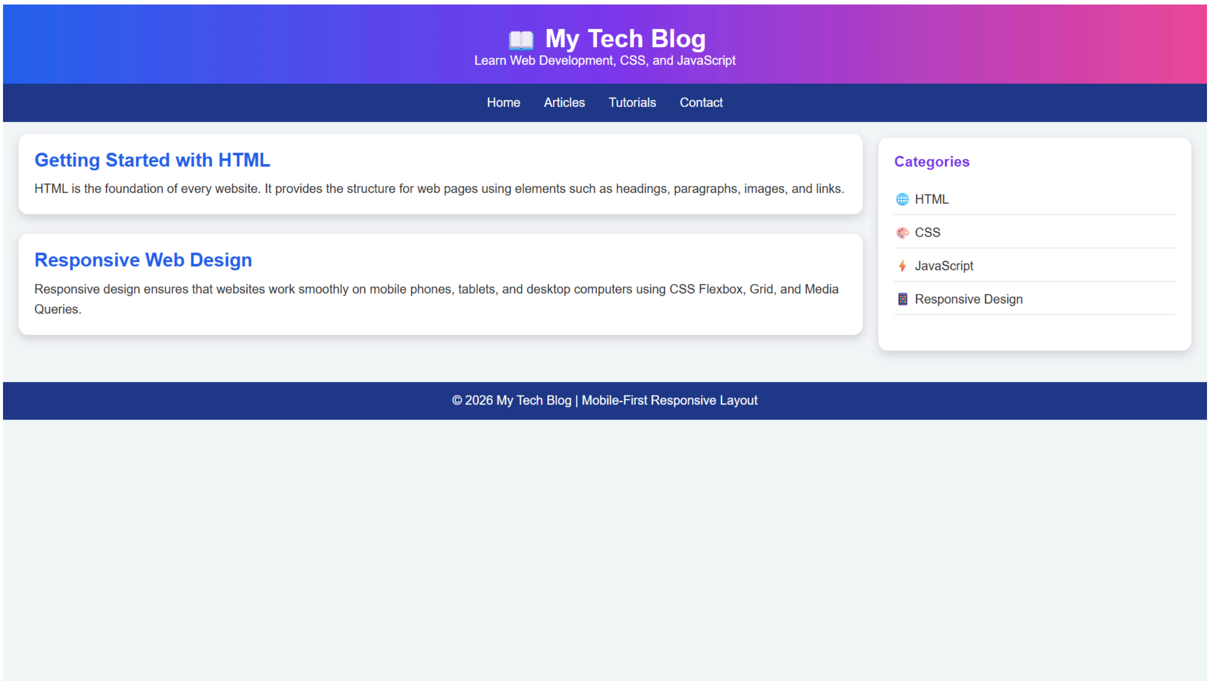
- Start with a mobile layout.
- Use Flexbox.
- Apply clean typography and spacing.

(c) Scaling

- Use media queries.
- Display sidebar beside content on larger screens.
- Convert navigation to horizontal layout.

(d) Evaluation

- Improves readability.
- Better accessibility.
- Works well across all screen sizes.



Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.
(a) Identify appropriate JavaScript events for menu interaction.
(b) Design event handling logic for menu toggle functionality.
(c) Implement interaction for user actions (click/hover).
(d) Evaluate performance and suggest improvements.

Ans:(a) JavaScript Events

- click
- mouseover
- mouseout

(b) Event Handling Logic

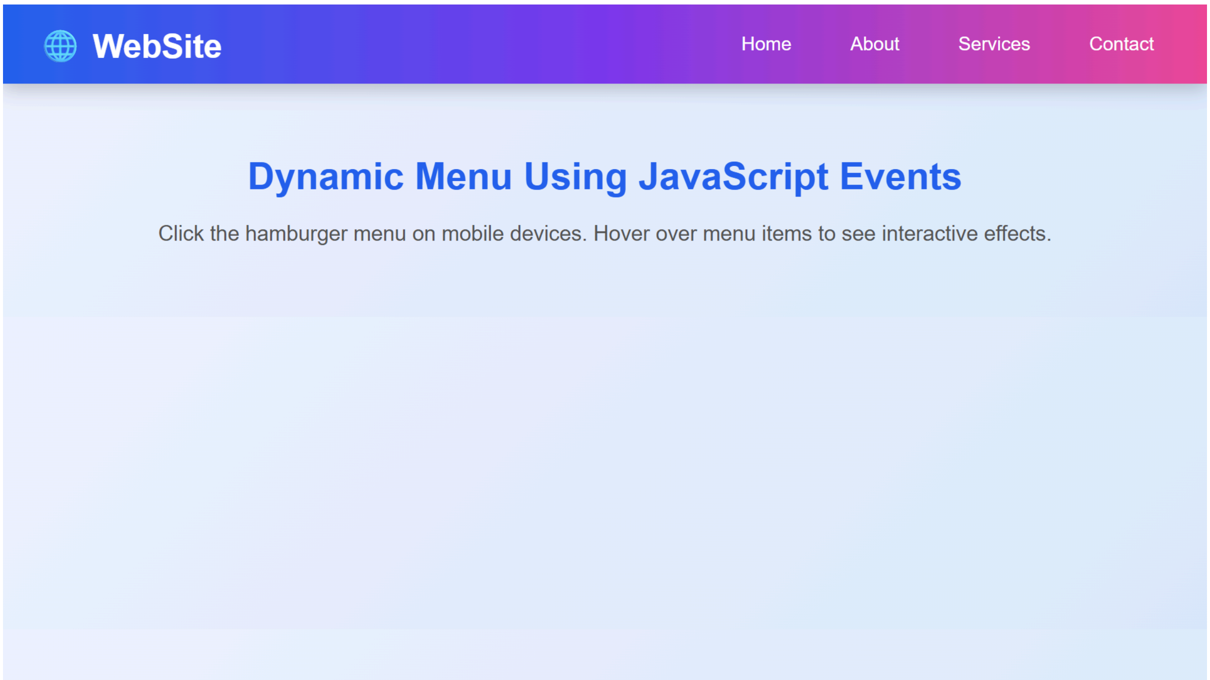
- Detect menu button click.
- Show or hide the navigation menu.
- Update menu state dynamically.

(c) User Interaction

- Click to open/close the menu.
- Hover to highlight menu items.
- Provide smooth visual feedback.

(d) Evaluation

- Improves navigation experience.
- Responsive on mobile devices.
- Use event delegation and optimized event listeners for better performance.



Code of Conduct:

I <u>G.MOKSHAGNA(192411160)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.MOKSHAGNA

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts

		requirements and concepts			
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation